Information Diffusion in Social Network

(SNA – B Tech III CSE)

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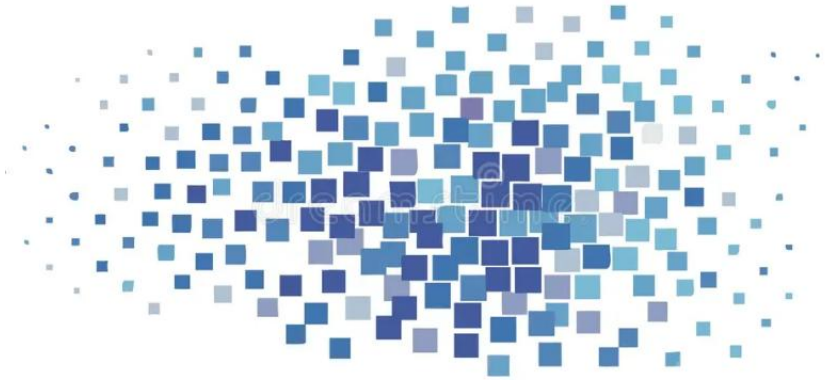

prologue

- Why do some ideas go viral and others die, even if they're equally good?"
- "Because networks; (not content) decide."

Questions and Roadmap..

- Questions –
- “How do we measure who matters?”
- “Which nodes accelerate diffusion?”
- “How do we design interventions?”
- Roadmap --
- Information diffusion concepts
- Network structure effects
- Echo chambers
- Key takeaways
- **Let’s first understand *how things spread***

Information Diffusion

- Information diffusion is the process by which ideas, behaviors, beliefs, or content propagate through a network via social connections.
- It’s a piece of information that flows from one individual or community to another in a network. It involves a source, a medium (e.g., social media), and receivers. Driven by user interactions, it acts as a, powerful tool for viral marketing, news dissemination, and **opinion formation.**
- Core elements are:
- Nodes = individuals / accounts / agents
- Edges = communication / observation / influence channels
- State = unaware, aware, adopter, sharer, resistant
- Used for: Identifying key influencers, understanding community dynamics, predicting trends, and controlling misinformation.

Diffusion

- **Mechanisms:** Information spreads through "contagious" processes, similar to diseases, with rapid, wide-scale propagation across platforms like Facebook and Twitter.
- **Network Structure:** The topological structure (nodes and edges) dictates the speed and reach, with "weak ties" often acting as bridges for spreading new information across different groups.

Concepts and Definitions

1. Information flow (message transmission)

- The **movement of a message** from one node to another through network connections, **regardless of whether it changes beliefs or behavior**.
- e.g. You see a tweet → information has flowed to you, even if you ignore it

2. Influence (change in belief/behavior)

- A **change in an individual's beliefs, attitudes, or intentions** resulting from exposure to others in the network, **with or without immediate action**.
- e.g. After seeing repeated posts, you start thinking a vaccine might be risky (belief changes), ~~but you still get vaccinated…~~

Cont.

3. Adoption (observable behavioral change)

- A **measurable action** indicating that the individual has acted on the information or belief. //A **verifiable action** in which an individual **implements or acts upon the information or belief**
- e.g., purchase, vote, share, install, participate
- **Information Flow → may lead to Influence → may lead to Adoption**
- But:
- Information can flow without influence
- Influence can occur without adoption
- Adoption is the strongest, observable outcome
- **Flow = exposure → Influence = internal change → Adoption = external action**

SI Model (Susceptible–Infected)

- Nodes are either Susceptible or Infected
- **States**
- **Susceptible (S):** Has not heard the information
- **Infected (I):** Has heard the information and can transmit it
- **Dynamics**
- Every time an infected node interacts with a susceptible neighbor,
  → there is a probability *p* that the neighbor becomes infected.
- There is **no forgetting** and **no resistance**. Once infected, always infected
- **What this models well**
- Breaking news, Viral memes, Awareness campaigns, Rumors that don't die quickly
- **Cache** - SI assumes memory is permanent and attention never fades—which is almost never true in real life

SIR Model (Susceptible–Infected–Recovered)

- **States**
- **Susceptible (S):** Not yet exposed
- **Infected (I):** Currently sharing / engaged
- **Recovered (R):** No longer interested / resistant / fatigued
- **Two probabilities now exist**
- Transmission probability (β)
- Recovery probability (γ)
- Recovered does NOT mean cured. It can mean: Lost interest, Fact-checked and rejected, Overloaded / attention saturated, Moved on to another topic
- Nodes can become resistant / lose interest
- Model epidemics, attention decay
- **E.g.** News cycle: people talk for 2–3 days, then forget
- Virality is a race between transmission and forgetting

Threshold Models

- Key idea: People do not adopt because of one exposure,
  They adopt when *enough people around them* have adopted.
- Each node has an adoption threshold θ
- E.g.: Alice adopts when ≥ 20% of friends adopt while Bob adopts when ≥ 60% of friends adopt
- It models peer pressure, conformity, norm formation, protest participation
- Cascade logic - A few early adopters trigger neighbors; Which triggers their neighbors; Which can cause a global cascade
- Real-world examples:
- Joining protests, Adopting technologies, Switching to WhatsApp when "everyone else is there", Wearing certain fashion, Buying luxury goods
- Threshold models explain why people say: I'll join only if others do first

Summary

- **SI model**
- Assumes people adopt after one exposure
- Good for simple awareness
- **SIR model**
- Adds forgetting / fatigue
- Models attention decay
- **Threshold model**
- Adds social dependence
- Models conformity, norms, peer pressure
- **But wait… all these models assume everyone is equal!!!!**

What these models still ignore?

- assumes homogeneous nodes
- ignores network structure
- ignores who is important
- This motivates centrality next.